Payal Mohapatra

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Research Interests: Robust and Efficient Machine Learning for Multimodal Time-Series (Audio, Speech, Wearables)

- **Robustness**: Distribution shift; Learning with incomplete modalities; Modality heterogeneity
- **Human-Centric Learning**: Metrics for subjective targets (auditory attention, fatigue); Data-efficient learning (pretraining new foundation models (FM), cross-modal alignment to LLMs/ FMs) ; Temporal and Mechanistic Interpretability (xAI)
- **Efficient On-Device Intelligence:** Sparse attention; Resource-aware inference acceleration

Education

Northwestern University — **Evanston, USA**
Ph.D. in Computer Engineering — *Sept 2021 – Early 2026 (Expected)*
Robust and Efficient Machine Learning for Time Series
Advisor: Dr. Qi Zhu | GPA: 3.96/4.0

Indian Institute of Technology Madras — **Chennai, India**
M.S. (by Research) in Electrical Engineering — *2015 – Nov 2017*
Algorithms and Systems for Wearable Health Monitors
Advisor: Dr. Mohanasankar Sivaprakasam | GPA: 9.31/10.0

Madras Institute of Technology, Anna University — **Chennai, India**
B.E. in Electronics and Instrumentation — *2011 – June 2015*
GPA: 9.36/10.0

Selected Experience

Industry

Mitsubishi Electric Research Lab (MERL) — **Boston, USA**
Part-time Student Researcher / Research Scientist Intern — *June 2025 – Dec 2025*
Vital Signs Estimation from Video Using Computer Vision and AI
Host: Tim K. Marks | Collaborators: Suhas Lohit, Hassan Mansour

- Built vision-based blood pressure monitoring system using custom camera, facial landmarking, and time-series extraction.
- Proposed signal-quality-based facial ROI construction with linear-phase filtering for pulse-transit-time feature extraction.
- Led end-to-end project: designed ML architecture, conducted IRB-approved data collection (8 subjects across 2 weeks), validated BP–PTT correlation, and built interactive visualizations; supporting large-scale publication efforts.

Meta Reality Labs — **Redmond, USA**
Part-time Student Researcher / Research Scientist Intern — *June 2024 – Dec 2024*
Behavior-driven Localization of Conversation-relevant Acoustic Zones Using Smartglasses
Host: Morteza Khaleghimeybodi | Collaborators: Ali Aroudi, Calvin Murdock, Buye Xu, Anjali Menon

- Formulated and developed an efficient sequential deep learning framework with hierarchical sensor fusion to predict acoustic zones of interest using IMUs.
- Built first proof-of-concept: Preprocessed and extracted head orientation features from IMU data across 70+ subjects collected using Aria glasses; developed LSTM with self-attention model for identifying the number of conversation partners (74% accuracy) and their locations.
- Achieved 35% improvement in localization performance over statistical and general-purpose time-series models for acoustic zone prediction.
- **Impact:** Contributed to a patent filing and authored a technical manuscript (under submission).

Meta Reality Labs — **Redmond, USA**
Part-time Student Researcher / Research Scientist Intern — *June 2023 – Dec 2023*
Non-Verbal, Discreet Communication Methods for Smartglasses
Host: Morteza Khaleghimeybodi | Collaborators: Ali Aroudi, Anurag Kumar

- Led end-to-end project: Designed IRB-approved experimental protocol, collected data from 21 subjects for teeth-click detection using contact microphones on smartglasses (Khadas VIM3 Amlogic A311D), and built complete ML pipeline.
- Developed lightweight Acoustic Event Detection model using broadcasted residual network, achieving 4% improvement over baseline.
- **Impact:** Deployed system in PyTorch/PyAudio on smartglasses hardware; demonstrated at internal Meta symposium with 88% positive reception (50+ participants rated willingness to adopt $\geq$3/5); authored technical white paper.

Analog Devices Inc. — **Bangalore, India**
Senior Design Engineer | Design Verification Engineer — *Nov 2017 – Sept 2021*
Digital Design and Formal Verification for ASICs

- Designed and verified (UVM and formal property verification using Cadence tools) digital subsystems for ASICs–audio noise cancellation, industrial Ethernet switches (IEEE 802.1AS time-sensitive networking), and ultrasound fingerprint sensing.
- Contributed to two chip tapeouts and post-silicon validation; presented Ethernet subsystem verification at global conference (GTC'2020); received Spot Award (top 1% globally) for methodology.

Academia

Northwestern University — **Evanston, USA**
Graduate Research Assistant — *Sept 2021 – Current*

- Developed ML frameworks for heterogeneous multimodal-sensing applications.
- Advanced speech-recognition technologies for minority and atypical users (ranging from disfluent to unvoiced EMG-based speech) in collaboration with Worcester Polytechnic Institute and Amazon Lab126.
- Led wearable systems research for operator fatigue and safety monitoring in manufacturing factory floors in collaboration with Boeing, John Deere, MxD, and University at Buffalo.
- **Impact:** First-authored 10 publications at top-tier venues (NeurIPS, ACL, Interspeech, ICASSP, TMLR, PNAS Nexus). Led three factory-floor demonstrations at Boeing and John Deere, featured in 5+ news outlets. Mentored 8+ students and contributed to 4 grant proposals.

Indian Institute of Technology Madras — **Chennai, India**
Graduate Research Assistant — *2015 – 2017*
Algorithms and Systems for Wearable Health Monitors

- Developed a custom optical sensor board for optimal signal quality in users with varying skin tones and validated it for pulse-rate variability and heart-rate (HR) estimation across 20 subjects.
- Developed a normalized least-mean-squared adaptive filtering-based motion-artifact rejection algorithm for real-time HR estimation.
- **Impact:** Authored three technical manuscripts (1 journal and 2 conferences including a best paper award at IEEE WinTechCon'18); demonstrated the overall system at the Healthcare Innovation Centre (HTIC), IITM Research Park.

Skills

Deep Learning: Transformers (`HuBERT, Wav2vec, iTransformer, Informer, Chronos`), LLMs (`Llama2, Llama3`), Self-Supervised Learning (`SimCLR, SimSiam`), Sparse/Window/Grouped-Query Attention, Time-Series Interpretability (`Captum`), Sequence Models (`LSTM, RNN, Mamba`), MoE, Domain Generalization (`DomainBed, BCResnet, AdaTime`), Facial Landmarking (`LuVLII, LipROI`), Multimodal Learning, Pruning, Distillation

Frameworks: PyTorch, Huggingface, Librosa, Pyaudio, OpenCV, TensorFlow, Scikit-learn, Pandas, LoRA, PyTorch DDP

Modalities: Audio, Speech, Vision, Surface Electromyography (sEMG), Contact Microphone, (remote) Photoplethysmography (PPG), Electrocardiogram (ECG), Electroencephalogram (EEG), Inertial Measurement Units (IMUs)

Programming Languages: Python (Pytorch, Scipy, Pandas, Tensorflow, plotly), C++, Shell, Perl, Verilog, System Verilog, Universal Verification Methodology (UVM)

Tools: Jetson, MAX78000, Khadas Vim3 (AI accelerators), LabVIEW, MATLAB, Spice Simulation, Google Datastudio, Data management (`PostgreSQL, Zenodo, Dataverse`), PCB Design, Xilinx Vivado, Git, Project management (`JIRA, monday.com`)

Misc.: Signal Processing (`Hilbert Transform, EMD, STFT, MFCC, Adaptive LMS Filtering`), Human-Subjects Research (`IRB, Study Protocol Design`)

Publications

In Conference Proceedings

2025: **Payal Mohapatra**, Yueyuan Sui, Akash Pandey, Stephen Xia, and Qi Zhu. "MAESTRO: Adaptive Sparse Attention and Robust Learning for Multimodal Dynamic Time Series." In **NeurIPS**, 2025. **(Spotlight, top 3.1%)**

2025: **Payal Mohapatra***, Akash Pandey*, Xiaoyuan Zhang*, and Qi Zhu. "Can LLMs Understand Unvoiced Speech? Exploring EMG-to-Text Conversion with LLMs." In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics* (**ACL**)*: Main Conference*, 2025.

2025: Akash Pandey*, **Payal Mohapatra***, Wei Chen, Qi Zhu, and Sinan Keten. "TimeSliver: Interpretable Time-Series Classification using Symbolic-Linear Representation." *Under review*, 2025.

2025: Yueyuan Sui*, **Payal Mohapatra***, Qi Zhu, and Stephen Xia. "MOSAIC: A Robust Learning Framework with Modality-Aware Adaptive Pruning for Multimodal Dynamic Time Series." *Under Review*, 2025.

2025: Lixu Wang, Bingqi Shang, Yi Li, **Payal Mohapatra**, Wei Dong, Xiao Wang, and Qi Zhu. "Split Adaptation for Pre-trained Vision Transformers." In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (**CVPR**), 2025.

2024: **Payal Mohapatra***, Shamika Likhite*, Subrata Biswas, Bashima Islam, and Qi Zhu. "Missingness-resilient Video-enhanced Multimodal Disfluency Detection." In **Interspeech**, 2024.

2023: **Payal Mohapatra***, Akash Pandey*, Sinan Keten, Wei Chen, and Qi Zhu. "Person Identification with Wearable Sensing Using Missing Feature Encoding and Multi-Stage Modality Fusion." In **ICASSP** 2023-2023 *IEEE International Conference on Acoustics, Speech and Signal Processing*. IEEE, 2023. **(3rd Place, Signal Processing Grand Challenge)**

2023: **Payal Mohapatra**, Bashima Islam, Md Tamzeed Islam, Ruochen Jiao, and Qi Zhu. "Efficient Stuttering Event Detection Using Siamese Networks." In **ICASSP** 2023-2023 *IEEE International Conference on Acoustics, Speech and Signal Processing*. IEEE, 2023.

2023: **Payal Mohapatra***, Akash Pandey*, Yueyuan Sui*, and Qi Zhu. "Effect of Attention and Self-Supervised Speech Embeddings on Non-Semantic Speech Tasks." In *Proceedings of the 31st ACM International Conference on Multimedia* (**ACM MM** '23), 2023. **(Top performer, ComParE challenge)**

2022: **Payal Mohapatra**, Akash Pandey, Bashima Islam, and Qi Zhu. "Speech Disfluency Detection with Contextual Representation and Data Distillation." In *Proceedings of the 1st ACM International Workshop on Intelligent Acoustic Systems and Applications*, 2022. (co-located with MobiSys'22)

2017: **Payal Mohapatra**, SP Preejith, and Mohanasankar Sivaprakasam. "A Novel Sensor for Wrist Based Optical Heart Rate Monitor." In *2017 IEEE International Instrumentation and Measurement Technology Conference* (**I2MTC**). IEEE, 2017.

Journal Articles

2025: **Payal Mohapatra**, Lixu Wang, and Qi Zhu. "Phase-driven Domain Generalizable Learning for Nonstationary Time Series." ***Transactions on Machine Learning Research***, 2025. | Time Series for Health (TS4H) Workshop, **NeurIPS** 2025 **(Spotlight, top 9.4%)**

2024: **Payal Mohapatra***, Vasudev Aravind*, Marisa Bisram, Young-Joong Lee, Hyoyoung Jeong, Katherine Jinkins, Richard Gardner, Jill Streamer, Brent Bowers, Lora Cavuoto, Anthony Banks, Shuai Xu, John Rogers, Jian Cao, Qi Zhu, and Ping Guo. "Wearable Network for Multilevel Physical Fatigue Prediction in Manufacturing Workers." ***PNAS Nexus***, volume 3, 2024.

2025: **Payal Mohapatra**, Calvin Murdock, Ishwarya Ananthbhola, Ali Aroudi, Anjali Menon, Buye Xu, and Morteza Khaleghimeybodi. "Towards Localizing Conversation Partners using Head Motion." *Under review*, 2025.

2018: **Payal Mohapatra**, Preejith Sreeletha Premkumar, and Mohanasankar Sivaprakasam. "A Yellow–Orange Wavelength-based Short-term Heart Rate Variability Measurement Scheme for Wrist-based Wearables." ***IEEE Transactions on Instrumentation and Measurement***. IEEE, 2018.

Patent

2025: *Systems and Methods for Identifying Zones of Audio Interest via Motion Sensors.* Filed: Jan 2025. Provisional Patent Application No. 63/746,543. Assignee: Meta Platforms, Inc. Inventors: **Payal Mohapatra**, Morteza Khaleghimeybodi, Ali Aroudi, Anjali Menon, Calvin Murdock, Buye Xu.

Preprint Articles

2024: **Payal Mohapatra**, Ali Aroudi, Anurag Kumar, and Morteza Khaleghimeybodi. "Non-verbal Hands-free Control for Smart Glasses Using Teeth Clicks." *arXiv preprint arXiv:2408.11346*, 2024.

Selected Honors

2025: **Terminal Year Fellowship**, Northwestern University.

2024: **EECS Rising Stars**, Massachusetts Institute of Technology (selected from top 19% of applicants nationwide).

2023: **Top-3 Performer**, ACM Multimedia Computational Paralinguistics Challenge (ComParE) (among 20+ international teams).

2023: **Top-3 Performer**, e-Prevention Challenge: Person Identification and Relapse Detection from Biosignals, ICASSP (among 15+ international teams).

2022: Travel Grant (1000 USD), ACM MobiSys Conference.

2022: CRA-WP Career Mentoring Workshop.

2021: **Best Research Video Award**, Design Automation Conference Young Fellowship (DAC YF).

2019: **Spot Award**, Analog Devices Inc. for scalable Ethernet switch verification (awarded to <1% of ADI employees globally).

2018: **Best Paper Award**, IEEE WinTechCon Conference.

2017: **Winner**, Make-in-India Anveshan Design Challenge, Analog Devices Inc.

2015-2017: Research Fellowship, Government of India (top 2% of applicants nationally).

2011-2015: National Merit Scholarship, Government of India.

Invited Talks

2025: Emerging Technology Track, IEEE Realtime Communications Workshop, Illinois Institute of Technology, Chicago.

2025: Computation and Data Exchange (CoDEx) 2025, Northwestern University.

Teaching Assistantship

Winter 2017: EE5400: Analog and Digital Circuits — IIT Madras

Fall 2016: EE5401: Measurements and Instrumentation — IIT Madras

Winter 2016: EE3006: Principles of Measurement — IIT Madras

Grant Writing Experience

Contributed to proposal writing and preliminary technical analyses for the following grants.

2025: Meta – Motor Learning & Neuromotor Ethics, PI: Qi Zhu, Stephen Xia, Maia Jacobs

2025: Amazon Research Award – Think Big, PI: Qi Zhu, Ping Guo

2025: Samsung Research America – Digital Health, PI: Qi Zhu, Ping Guo

Professional Services

2026: Organizing Committee, The Ninth International Workshop on Computer Vision for Physiological Measurement (proposed for CVPR 2026)

2025: Technical Program Committee, International Workshop on Intelligent Acoustic Systems and Applications

Reviewer

2025: IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)

2025: IEEE Transactions on Circuits and Systems for Artificial Intelligence (TCAS-AI)

2025: IEEE Transactions on Automation Science and Engineering (TASE)

2025: IEEE International Workshop on Machine Learning for Signal Processing (MLSP)

2025: Workshop on Foundation Models for Structured Data (FMSD) at ICML

2025: Workshop on Time Series in the Age of Large Models (TSALM) at NeurIPS

2024, 2025: Annual Conference on Neural Information Processing Systems (NeurIPS)

2024, 2025: Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)

2024, 2025: IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)

2025, 2026: International Conference on Learning Representations (ICLR)

2024: IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

2024: Asia and South Pacific Design Automation Conference (ASP-DAC) (secondary reviewer)

2023: IEEE Internet of Things Journal (IoT-J)

2023: International Conference on Embedded Software (EMSOFT) (secondary reviewer)

2023: ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS) (secondary reviewer)

2022: International Conference on Networking, Systems and Security (NSys) (secondary reviewer)

2023: Book: *Why Does Math Work... If It's Not Real?*, Cambridge University Press

Mentoring

2025: Xiaoyuan Zhang, Haodong Yang (MS, Computer Engineering)

2024: Talia Ben-Naim (MS, Computer Engineering → Embedded Software Engineer, Medtronic, Boston), Brooks Hu (Undergraduate, Computer Engineering), Mark Zhang (MS, Mechanical Engineering)

2023: Yueyuan Sui (MS → PhD, Northwestern University), Shamika Likhite (MS → Software Engineer, SpeechAce), Kiva Joseph (Undergraduate, Computer Engineering), Jonathan Li Chen, Ben Forbes, Justin Lau (Undergraduates, Mechanical Engineering)

2022: Devashri Naik (MS → PhD, University of Illinois at Chicago), Jinjin Cai (MS → PhD, Purdue University)